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Paper Code : ESC 301/ESC301/PC-ROB301, Analog and Digital Electronics
UPID : 003442

CS/B.TECH(N)/ODD/SEM-3/3442/2023-2024/1018

Time Allotted : 3 Hours

Full Marks : 70

The Figures in the margin indicate full marks.

Candidate are required to give their answers in their own words as far as practicable

Group-A (Very Short Answer Type Question)

1. Answer any ten of the following :

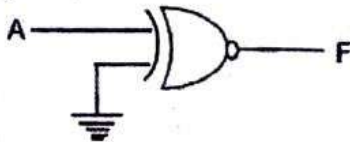
[1 x 10 = 10]

- (i) What will be the minimal product-of-sums function described by the K-map given in Fig.

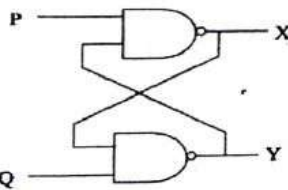
AB	00	01	11	10
C				
0	1	1	0	0
1	0	0	0	0

- (ii) Arrange the Classes of power amplifier according to their efficiency (low to high)
Class A , Class B , Class C , Class AB

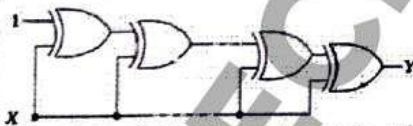
- (iii) What will be the output of the logic gate in figure given below :



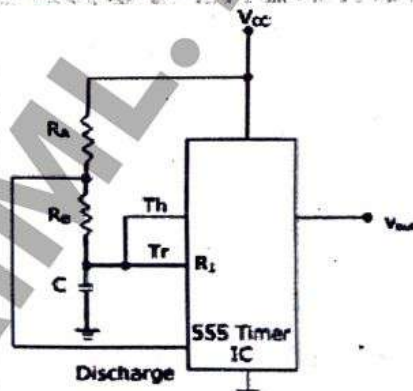
- (iv) In the latch circuit shown, the NAND gates have non-zero, but unequal propagation delays. The present input condition is: $P = Q = "0"$. If the input condition is changed simultaneously to $P = Q = "1"$, what will be the outputs X and Y now ?



- (v) If the input to the digital circuit (in the figure) consisting of a cascade of 20 XOR-gates is X. What will be the output Y ?

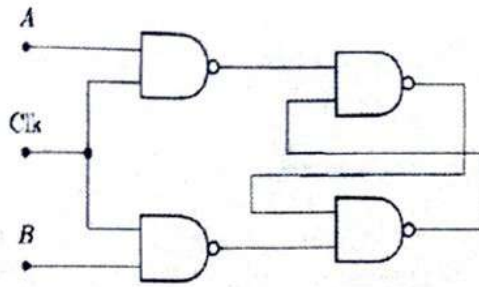


- (vi) The circuit of following figure shows a IC 555 Timer connected as an Astable multivibrator. The value of the capacitor C is 10 nF. Find the values of the resistors R_A and R_B for a frequency of 10 kHz and a duty cycle of 0.75 for the output voltage.



- (vii) What will be the minimum number of 2 to 1 multiplexers required to realize a 4 to 1 multiplexer ?

- (viii) Consider the given circuit. Explain whether , the race around will occur or not



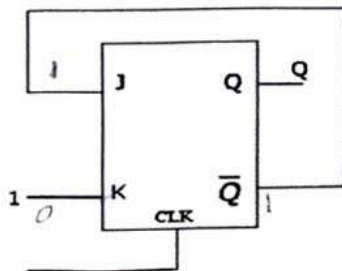
- (IX) Which A/D converter is used for Hum rejection ?
- (X) The a.c. output power of a Class B push-pull power amplifier is 10 watt. What will be the d.c. input power drawn from power supply when the efficiency of the is maximum .
- (XI) What is octal value of $(2^6)_{10}$?
- (XII) Statement 1: Astable Multivibrator can be used for generating Square Wave.
Statement 2: Bistable Multivibrator can be used for storing binary information.
Please state whether both the statements are true or false.

Group-B (Short Answer Type Question)

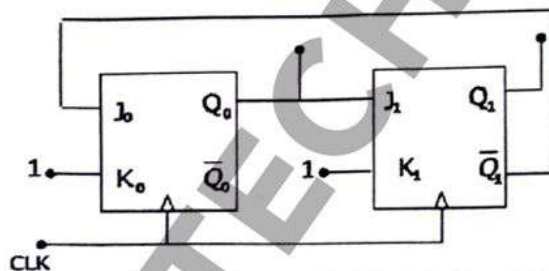
Answer any three of the following :

[5 x 3 = 15]

2. In a JK flip-flop, we have $J=Q'$ and $K=1$ (see figure). Assuming the flip-flop was initially cleared and then clocked for 6 pulses, What will be the sequence at the Q output ? [5]



3. Derive the maximum efficiency of a class B amplifier [5]
4. Explain the operation of transformer coupled Class A amplifier [5]
5. Find out what will be the modulus of this counter? [5]



6. Design a Full subtractor (X,Y and Borrow) with 4 : 1 MUX [5]

Group-C (Long Answer Type Question)

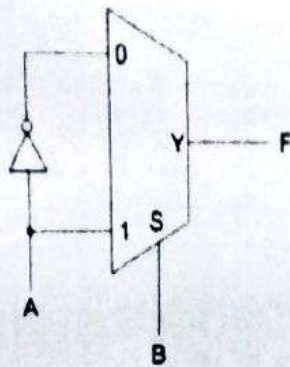
Answer any three of the following :

[15 x 3 = 45]

7. (a) Find the out expression for the Karnaugh Map shown below [5]

CD \ AB	00	01	11	10
00	0	0	0	0
01	1	0	0	1
11	1	0	1	1
10	0	0	0	0

- (b) Find the out F interms of A and B [3]



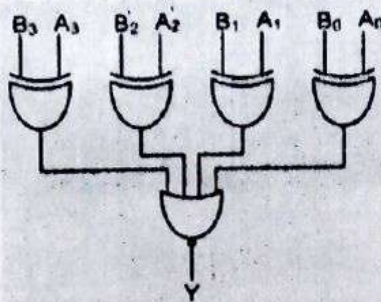
(c) Find the output expression F for the following Karnaugh Map and realise it with logic gates

[7]

AB \ C	00	01	11	10
0	1	1		
1		1	1	

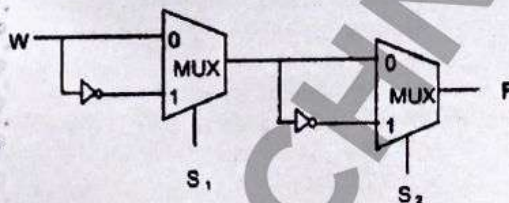
8. (a) A digital circuit which compares two numbers $A_3A_2A_1A_0$ (A) $B_3B_2B_1B_0$ (B) is shown in figure. Find the pair A, B to get output $Y = 0$

[5]



(b) Consider the multiplexer based logic circuit shown in the figure. Find the Boolean functions realized by the circuit.

[5]

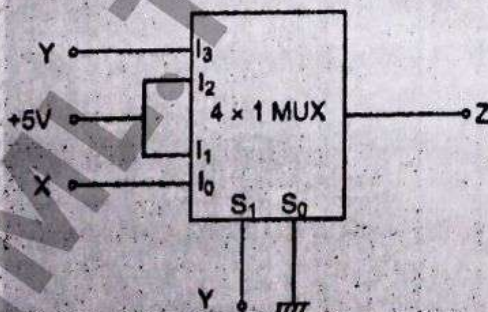


(c) A 4 bit modulo-16 ripple counter uses JK flip-flops. If the propagation delay of each FF is 50ns. Find the maximum clock frequency that can be used in this counter:

[5]

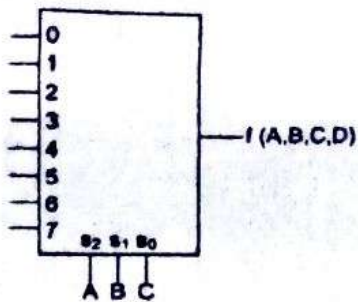
9. (a) Find the output Z in terms of X and Y

[8]



(b) A Boolean function $F(A,B,C,D) = \pi(1, 5, 12, 15)$ is to be implemented using an 8x1 multiplexer (A is MSB). The inputs ABC are connected to the select inputs $S_2S_1S_0$ of the multiplexer respectively.

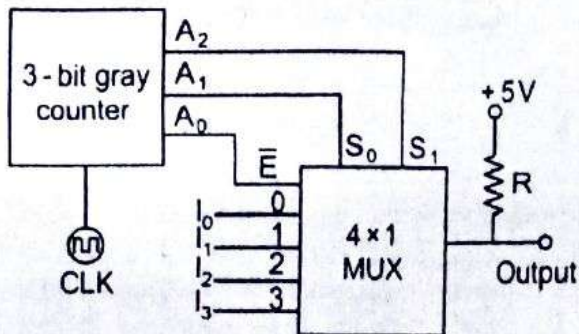
[7]



Find the correct inputs to pins 0, 1, 2, 3, 4, 5, 6, 7 in order.

10. (a) A 3-bit gray counter is used to control the output of the multiplexer as shown in the figure (A_2 is MSB and A_0 is LSB). The initial state of the counter is 000_2 . The output is pulled high. Find the sequence of the output of the circuit

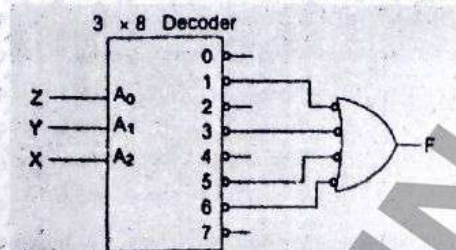
[8]



- (b) A 3 line to 8 line decoder, with active low outputs, is used to implement a 3-variable Boolean function as shown in figure:

[7]

Find the simplified form of Boolean function $F(A,B,C)$ implemented in 'Product of Sum' form.



11. (a) Digital input signals A, B, C with A as the MSB and C as the LSB are used to realize the Boolean function $F = m_0 + m_2 + m_3 + m_5 + m_7$, where m_i denotes the i th minterm. In addition, F has a don't care for m_1 .

[10]

Find the simplified expression for F

- (b) Find the prime implicants in the sum of products function $(X,Y,Z) = \sum(2,3,4,5)$

[5]

*** END OF PAPER ***